

Machine learning systems engineer building and optimizing production ML services and GPU infrastructure. I establish technical foundations on small teams and scale them through company growth.

TECHNICAL SKILLS

ML & inference: PyTorch, NVIDIA Triton Inference Server, TensorRT-LLM, CTranslate2, NVIDIA Riva

Infrastructure: Kubernetes (GKE), GCP, AWS, Terraform, Helm, ArgoCD, KEDA, GitHub Actions

Services & data: Python, asyncio, FastAPI, Pydantic, Temporal, gRPC, WebSockets, SQL (PostgreSQL, TimescaleDB)

EXPERIENCE

Abridge

Remote

Senior Machine Learning Engineer

Aug 2024 - Dec 2025

Machine Learning Engineer

Apr 2023 - Jul 2024

Joined a Series B healthcare startup as employee ~25 and ML engineer #2; company grew to 400+ employees

- Built and operated a multi-model ASR platform on GKE for models developed by the research team, separating preprocessing, inference, and postprocessing into Triton stages with tuned concurrency and parallel execution; processed **4 audio-hours per second** at daily peaks and up to 8 during backlog recovery.
- Migrated Whisper ASR from **2,000+ NVIDIA L4 GPUs** to **100-200 H100s** by replacing CTranslate2 with TensorRT-LLM and increasing batch sizes, serving the same production load at lower cost per audio-minute.
- Deployed real-time ASR on GKE and H100s using NVIDIA Parakeet and Riva, delivering live in-app transcription over WebSockets and gRPC.
- Partnered with Platform to establish shared GKE deployment practices, writing the company's first in-house Helm chart and helping move services from Cloud Functions and App Engine to GKE.
- Built a unified FastAPI interface across ML services, standardizing upstream integration, observability, and operational patterns.
- Introduced Temporal for complex asynchronous workflows and championed its adoption across **nearly all engineering teams**.
- Built monorepo development and release tooling for dozens of engineers, enabling cross-component changes in a single PR and reducing support requests for PRs and deployments.
- Mentored engineers in asynchronous Python and Helm deployment, enabling them to build and deploy new services independently.
- Led incident response for recurring egress port exhaustion and drove connection-reuse improvements across Python and JavaScript services.

Socure

Remote

Staff Data Engineer, Data Science Enablement & Operations

Jun 2022 - Jan 2023

Most senior engineer supporting data science teams with tooling and process

- Refined an internal batch API with service owners and shipped its Airflow integration.
- Generated a Python client for the customer-facing API and built reusable abstractions to simplify its use in Airflow pipelines.

Betterment

Remote

Lead Data Engineer

Jun 2021 - Dec 2021

Senior engineer on a centralized data engineering team owning databases and pipelines

- Standardized core Airflow reporting on Python, replacing R to simplify maintenance and ownership.
- Introduced Pydantic for AWS Lambda tasks.

EXPERIENCE CONTINUED

Everactive

Senior Staff Engineer, Cloud Platform

Staff Engineer, Data Science

Ann Arbor, MI

Jan 2021 - May 2021

Mar 2019 - Jan 2021

First data science hire at a Series B industrial IoT startup

- Designed a unified TimescaleDB/PostgreSQL platform for sensor readings and metadata, migrating **billions of rows online** from OpenTSDB/EMR and enabling interactive customer graphing.
- Flattened an overly nested sensor data model through further online migrations to improve TimescaleDB write performance.
- Served as technical lead for a company-wide API consolidation project, standardizing access to core sensor data across teams.
- Authored initial JSON Schema contracts for gateway-to-cloud data and secured senior-engineer buy-in on semantic versioning and compatibility requirements.
- Wrote job descriptions and conducted interviews for subsequent data science hires.

Methods Consultants

Data Science Engineer

Ypsilanti, MI

Mar 2016 - Feb 2019

First full-time hire and lead engineer at a boutique data science consulting firm

- Led development of an **ML investment platform** for the firm's largest client, combining proprietary signals and external financial data in ensembles of regression, random forests, gradient boosting, and neural networks.
- Built a backtesting platform and automated model evaluation, backtesting, and daily prediction workflows on Pachyderm and GKE.
- Analyzed billions of geolocation pings with R and OpenStreetMap to support expert-witness testimony in a trucking fraud case.
- Developed and taught a week-long R and SQL workshop for analysts at a large federal institution.
- Introduced version control with Git and GitLab and cloud computing with AWS and GCP, establishing foundations for the firm's engineering work.
- Hired and mentored junior data scientists.

University of Michigan School of Social Work

Data Analyst (part-time)

Ann Arbor, MI

Aug 2015 - Dec 2017

Industry consultant supporting a data-driven social work research lab

- Built predictive models to classify journal abstracts, redesigned pipelines for administrative data, and created reusable lab tooling.

Barracuda Networks

Business Intelligence Analyst

Performance Analyst/Software Engineer in Test

Ann Arbor, MI

Mar 2014 - Feb 2016

Aug 2012 - Mar 2014

QA engineer turned founding business intelligence analyst at a large software company

- Created data warehouses to provide the company's first historical view of critical production data.
- Managed server infrastructure and developed R tooling for cloud capacity planning.
- Identified performance regressions and worked with engineering to determine their causes.

EDUCATION

Michigan Technological University

Bachelor of Science in Mathematics, minor in Economics

2012